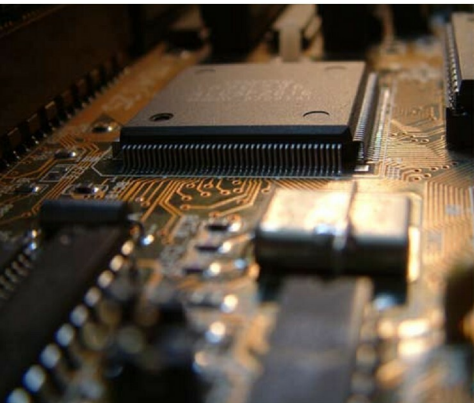


SEMICONDUCTOR ICs IN THE MAKING

Job prospects for electronics engineering graduates are bright in IC manufacturing companies. Take a look at what it takes to manufacture an IC



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Transistors are the fundamental units of all sophisticated chips (closely and highly integrated packages). According to Moore's law, the capacity of ICs almost doubles every two years. It has evolved currently to 45nm technology in the VLSI fabrication field. Nowadays, chips are designed with more than a million transistors in one square centimetre, i.e., a single transistor occupies about 100 nm.

Metal-oxide semiconductor field-effect transistors (MOSFETs) are superior to transistors because of their lower power consumption, higher switching speed, ease of fabrication and robustness. So let us take a MOSFET and see how it is fabricated by the diffused planar process used for large-scale IC fabrication.

Steps for manufacturing of MOS devices

These unipolar transistors have source (equivalent of emitter), drain (equiva-

lent of collector) and gate (equivalent of base) regions accessible from the top surface of the silicon substrate. Once the source, gate and drain have been formed in the substrate by diffusing appropriate impurities, aluminium or tungsten contacts can be added to these areas on the wafer's surface.

To be specific, the following steps are performed for manufacturing a MOS device:

1. Substrate preparation
2. SiO_2 insulating layer deposition (initial oxidation)
3. Photolithographic process-1
 - (a) Application of photoresist layer-1
 - (b) Application of initial (layer-1) mask
 - (c) Oxide etching
4. Impurity predeposition
5. Impurity deposition and de-oxidation
6. Photolithographic process-2
 - (a) Application of photoresist layer-2
 - (b) Application of layer-2 mask
 - (c) Oxide etching
7. Impurity deposition and de-oxidation
8. Photolithographic process-3
 - (a) Application of photoresist layer-3
 - (b) Application of layer-3 mask
 - (c) Oxide etching
9. Metal deposition and evaporation
10. Wafer processing
11. Wafer testing
12. IC packaging
13. IC package testing